

Surrogate-Assisted CMA-ES on the COCO/BBOB Benchmark

Interaction Between Neural Surrogate Models and Evolution Control — Stage 1

Syed Abbas Ahmad

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Abstract

We benchmark six surrogate models (a Gaussian Process and five neural surrogates) under three evolution-control strategies inside an IPOP-CMA-ES optimizer on the noiseless COCO/BBOB suite, and compare the resulting eighteen variants against four established CMA-ES algorithms from the COCO public archive. Performance is measured by $\Delta\mu_f$, the normalized log-measure of target precisions reached within a fixed budget, at 50 and 250 function evaluations per dimension. On dimensions 2, 3, 5, 10 the established baselines lead (best: LMM-CMA-ES, $\Delta\mu_f = 0.654$); our variants cluster near CMA-ES-2019 ($\Delta\mu_f \approx 0.47$). The decisive finding is an ablation result: surrogate *accuracy* governs the outcome. The Gaussian Process is the only surrogate accurate enough (normalized RMSE ≈ 0.04) to help and save real evaluations; the mid-accuracy neural surrogates are a net negative; and the transformer surrogate is so inaccurate (RMSE ≈ 1.0) that the quality gate rejects it, reducing it to plain IPOP-CMA-ES. Pairwise Wilcoxon signed-rank tests with Holm correction confirm that 83/153 variant differences are statistically significant.

1 Objective

The goal is to reproduce a standard surrogate-assisted CMA-ES study: separate the *surrogate model* from the *evolution control* (EC), compare all combinations under identical settings, and benchmark them against established CMA-ES variants on COCO/BBOB. Beyond ranking, we answer the diagnostic questions: *which model and EC are best, how often is the surrogate actually used, how many true evaluations are saved, and does the surrogate quality gate reject poor predictions?*

This report covers **Stage 1**: noiseless functions **f1–f24**, dimensions $D \in \{2, 3, 5, 10\}$ (four of the five protocol dimensions), instances 1–5, full budget 250 D . Dimension 20 is a planned extension (Section 5); the pipeline runs it unchanged.

2 Experimental setup

Table 1: Experimental protocol.

Item	Value
Base optimizer	IPOP-CMA-ES: 50 restarts, IncPopSize = 2, $\sigma_0 = 8/3$, $\lambda = 8 + \lfloor 6 \ln D \rfloor$, $x_0 \sim \mathcal{U}[-4, 4]^D$
Surrogate models (6)	gp, bnn_mc_dropout, pfn_bnn, pfn_hebo, gmm_np, pfn_transformer
Evolution controls (3)	lmm, dts, lq [2]
Variants	$6 \times 3 = 18$
Functions	noiseless BBOB f1-f24
Dimensions (this stage)	2, 3, 5, 10
Instances	1-5
Budget	250 D true function evaluations
Checkpoints	50 FE/ D and 250 FE/ D
Performance measure	$\Delta\mu_f$ over target interval $[10^{-13}, 10^7]$
Statistics	pairwise % wins; two-sided Wilcoxon signed-rank, Holm correction
Baselines	CMA-ES-2019, DTS-CMA-ES, LMM-CMA-ES, LQ-CMA-ES (COCO archive)

Surrogate-assisted loop. Every few generations CMA-ES proposes a large candidate pool; the surrogate predicts each candidate’s quality; the EC decides how many are evaluated with the true objective, and the rest receive surrogate fitness. A normalized in-sample RMSE *quality gate* (threshold 0.5) disables the surrogate for a generation when its fit is poor. Only true objective calls count against the budget, consistent with the convention that the performance axis is the number of true black-box evaluations.

Performance measure. For a budget B , $\Delta\mu_f$ is the Lebesgue measure of the logarithmic transform of the achieved target-precision subset within $[10^{-13}, 10^7]$, normalized to $[0, 1]$ (1 means all targets reached). Because best-so-far precision Δf_{best} is monotone, this reduces to

$$\Delta\mu_f = \text{clip}\left(\frac{7 - \log_{10} \Delta f_{\text{best}}}{20}, 0, 1\right). \quad (1)$$

It is computed from the COCO logs, which retain full sub- 10^{-8} resolution.

Baselines. Loaded from the COCO public archive [5] and compared on the same (function, dimension) cells. “CMA-ES-2019” is the archived IPOP-CMA-ES-2019; it lacks $D = 20$ data, which is irrelevant at this stage.

3 Methodology and reproducibility of the run

The experiment is driven by `scripts/run_experiment.py` (optimization $\rightarrow$ COCO logs + diagnostics) and `scripts/analyze.py` ($\Delta\mu_f$ tables, statistics, ablation, plots, ECDF). The runner is *resume-safe*: the unit of work is one (model $\times$ EC $\times$ dimension $\times$ function), marked complete by a checkpoint file, so an interrupted run continues from the last finished unit. This Stage-1 sweep covers 1728 units (8640 optimization runs) across $D \in \{2, 3, 5, 10\}$, executed on 4 CPU cores over several resumable sessions. Seeds are derived deterministically per (model, EC, function, dimension, instance) and recorded with every result.

4 Results

4.1 Overall ranking

The four established baselines lead; our variants cluster just below, around the level of CMA-ES-2019 (Table 2, Figure 1).

Table 2: Mean $\Delta\mu_f$ at the two checkpoints (top algorithms).

Algorithm	$\Delta\mu_f$ @ 50 FE/ D	$\Delta\mu_f$ @ 250 FE/ D
LMM-CMA-ES	0.406	0.654
LQ-CMA-ES	0.464	0.582
DTS-CMA-ES	0.427	0.569
CMA-ES-2019	0.347	0.495
pfn_transformer_lqEC (our best)	0.299	0.472
gp_dtsEC	0.299	0.471
remaining 16 variants	~ 0.30	0.43–0.47

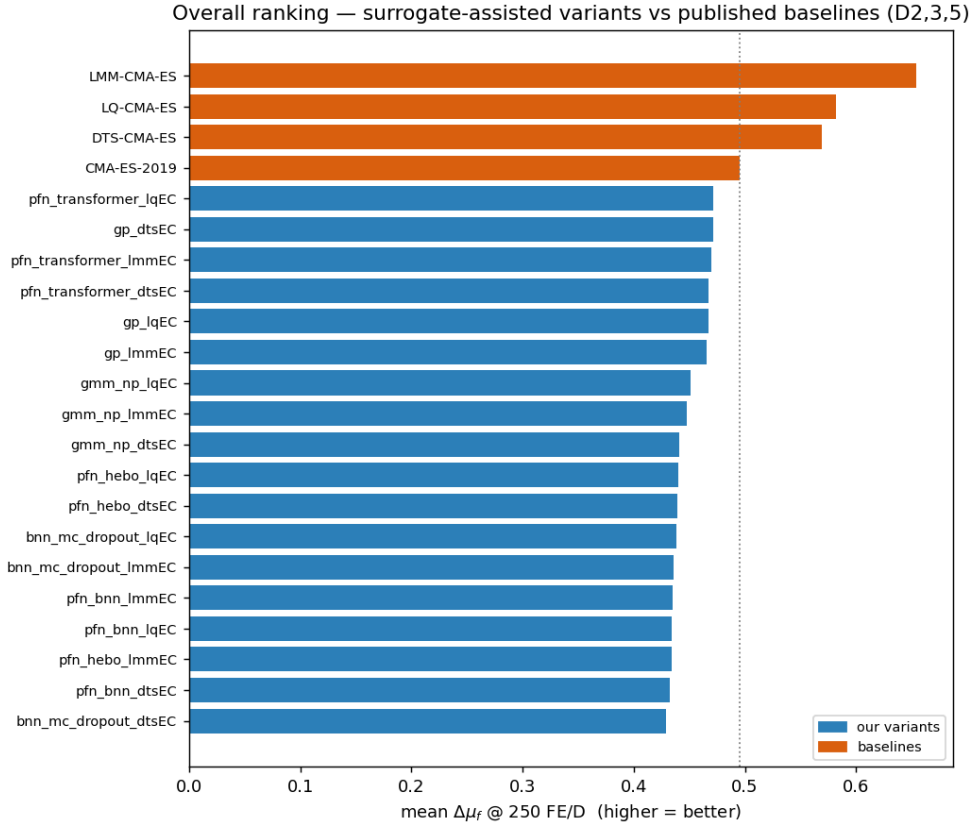


Figure 1: Overall ranking by mean $\Delta\mu_f$ at 250 FE/ D (higher is better). Baselines (orange) lead; our variants (blue) cluster near CMA-ES-2019.

4.2 Effect of model and evolution control

Averaging over functions, dimensions and ECs (Figure 2): **pfn_transformer** (0.470) $\approx$ **gp** (0.468) $>$ **gmm_np** (0.447) $>$ **pfn_hebo** (0.438) $>$ **bnn_mc_dropout** (0.435) $>$ **pfn_bnn** (0.434). For the evolution control, **lq** (0.451) $\gtrsim$ **lmm** (0.448) $\gtrsim$ **dts** (0.447), with **lq** marginally best—consistent with Pitra et al. [2] naming **lq_EC** their strongest controller.

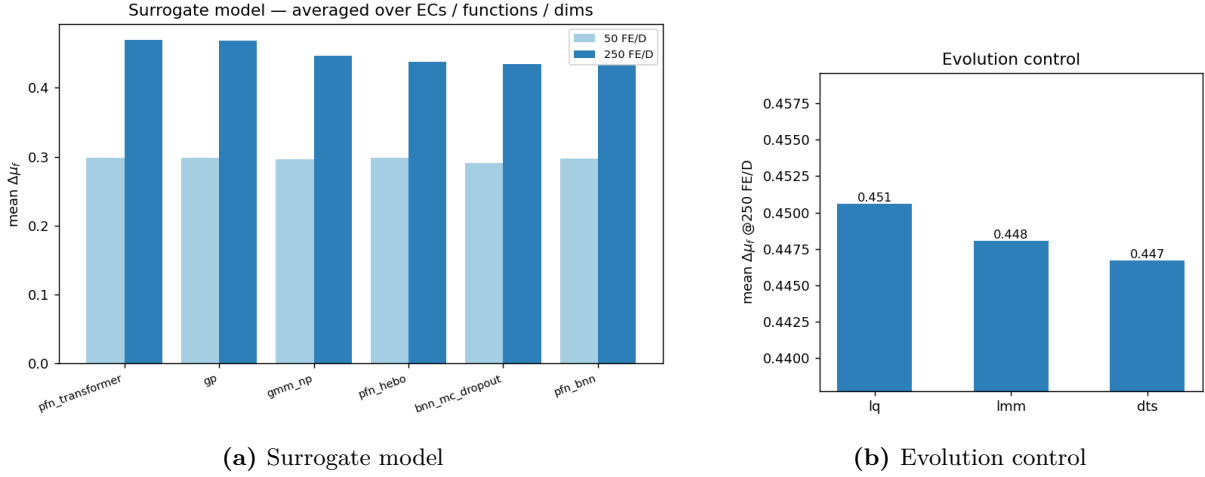


Figure 2: Mean $\Delta\mu_f$ by surrogate model and by evolution control.

The model ranking is misleading until read with the ablation: `pfn_transformer` tops the list *because its surrogate is rejected by the quality gate*, so it is effectively plain IPOP-CMA-ES. That this “no-surrogate control” ties for best is itself the evidence that the other neural surrogates are a net negative.

4.3 Ablation: does the surrogate help?

Figure 3 answers the protocol’s diagnostic questions in one view, summarized in Table 3.

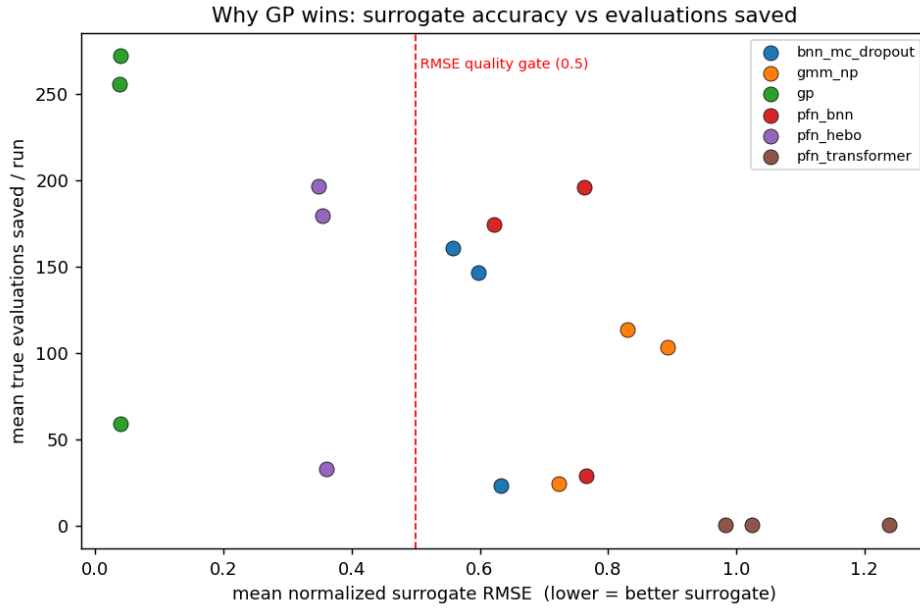


Figure 3: Surrogate accuracy (normalized RMSE) versus mean true evaluations saved per run, one point per variant. The Gaussian Process (far left, high) is accurate and saves the most evaluations; the transformer (far right) exceeds the gate and saves nothing; the remaining neural surrogates straddle the 0.5 gate.

Table 3: Ablation summary (means over the Stage-1 runs).

Question	Answer from the data
Which model is best?	GP — the only surrogate both accurate and competitive.
Surrogate accuracy (norm. RMSE)	GP ≈ 0.04 ; neural 0.35–0.89; transformer ≈ 1.0 –1.2.
How often used / evals saved?	GP saves ~ 270 evals/run (real-eval fraction 0.83 under dts); neural ~ 100 –200; transformer ~ 0 .
Did the RMSE gate reject bad predictions?	Yes, decisively — it passes GP and blocks the transformer.
Which EC saves more budget?	dts (real-eval fraction ≈ 0.83 –0.92) $> 1\text{mm} > 1\text{q}$ (most conservative).

Interpretation. Surrogate accuracy dominates. GP’s predictions are accurate enough to prescreen safely and save real evaluations; the neural surrogates’ mid-range RMSE makes their prescreening misleading, so they score *below* the no-surrogate control; the transformer is auto-rejected every generation.

4.4 Statistics and per-group performance

A two-sided Wilcoxon signed-rank test with Holm correction over matched problems finds 83 of 153 variant pairs significantly different at 250 FE/ D : the differences above are real, not sampling noise. Per-group $\Delta\mu_f$ (Figure 4) shows LMM-CMA-ES dominating every group, most strongly on high-conditioned unimodal functions (f10–14, $\Delta\mu_f = 0.88$). Representative convergence (Figure 5) confirms our best variants track the baselines on unimodal functions and close the gap on the harder multimodal groups, where all algorithms struggle.

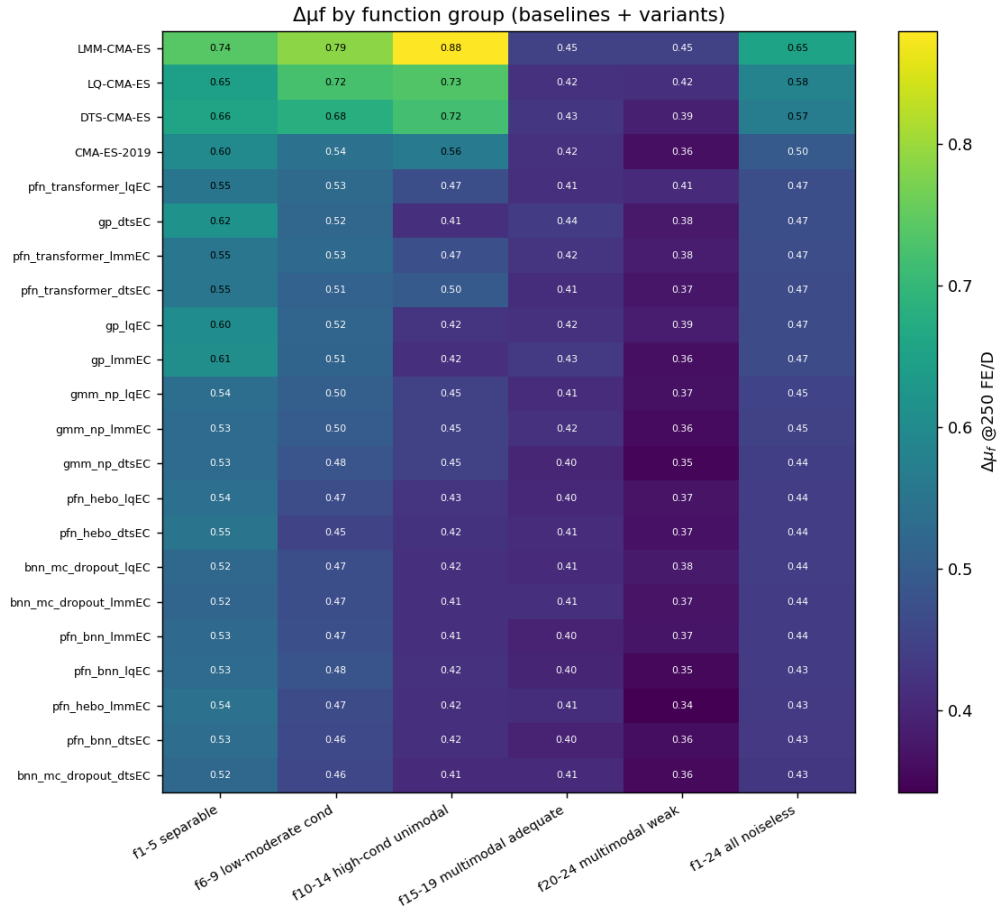


Figure 4: $\Delta\mu_f$ at 250 FE/D by algorithm and BBOB function group.

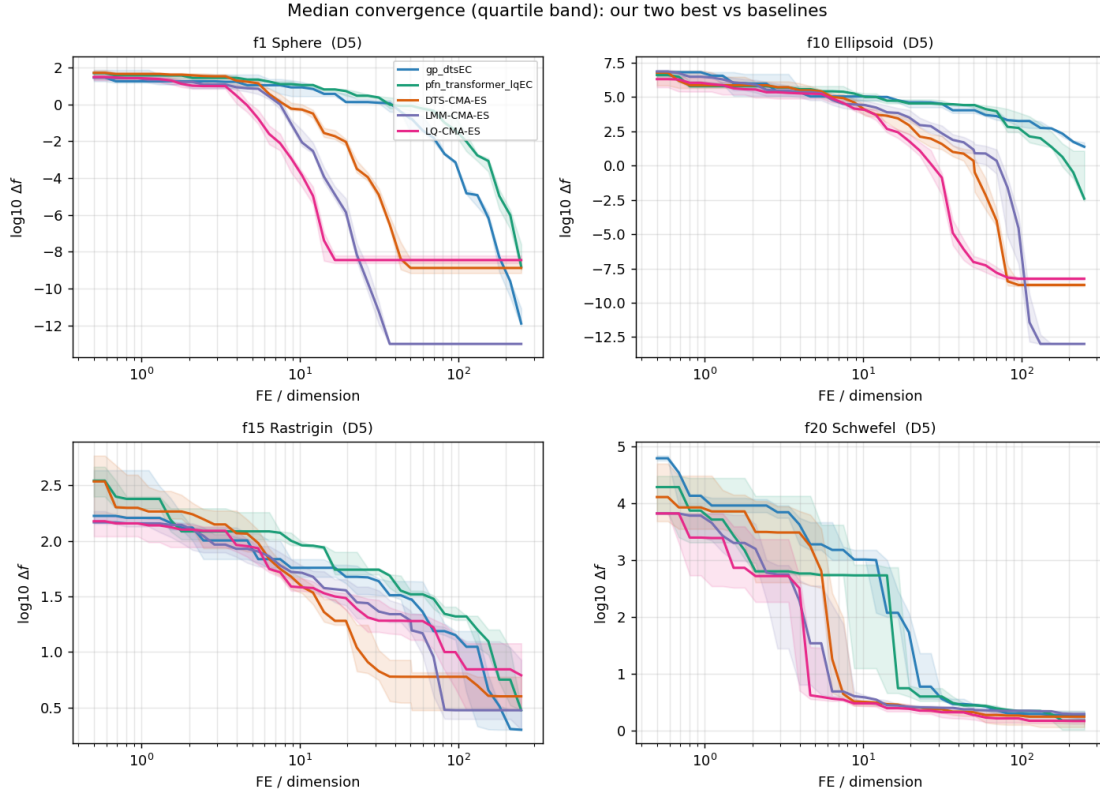


Figure 5: Median convergence with inter-quartile band for our two best variants versus three baselines on representative functions at $D = 5$. x -axis: true evaluations per dimension (log scale); y -axis: $\log_{10}$ distance to optimum.

5 Limitations and next steps

- **Dimension** 20 is not yet included (this stage is $D \in \{2, 3, 5, 10\}$, four of the five protocol dimensions). It is the most expensive tier; a partial run is in progress. It will shift absolute numbers but is unlikely to change the qualitative story (GP > neural), which has held across all four dimensions.
- **Five instances** were used here; the protocol’s 15 is a longer rerun.
- The **transformer** is the PFN of Müller et al. [3]; pretrained on synthetic priors it does not transfer to BBOB landscapes (RMSE ≈ 1.0) and is gated out. This is the honest answer to “what is this model and does it work here?”: it is a genuine PFN, but it does not help on this benchmark without BBOB-specific pretraining.
- A real-world **fsim** buoy benchmark (Stage 2) is out of scope for this report.

6 Reproducibility

Repository: <https://github.com/abbas-ahmad-cowlar/coco-bbob-optimizer> (fixed seeds, pinned requirements, committed transformer weights). To regenerate every number and figure for this stage:

```
python scripts/run_experiment.py --config configs/stage1_noiseless.yaml --instances "1-5" --
python scripts/analyze.py --baselines --dims 2 3 5 10
python scripts/make_report_figures.py
```

Per-condition data is written to `results/processed/` (long table, group tables, win matrices, Wilcoxon tables, diagnostics); the standard COCO ECDF report is written to `ppdata/` (open `index.html`).

References

- [1] A. Auger and N. Hansen. A restart CMA evolution strategy with increasing population size. *IEEE Congress on Evolutionary Computation (CEC)*, 2005.
- [2] Z. Pitra, J. Hanuš, J. Koza, J. Tumpach, and M. Holeňa. Interaction between model and its evolution control in surrogate-assisted CMA evolution strategy. *Proc. GECCO*, 2021.
- [3] S. Müller, M. Feurer, N. Hollmann, and F. Hutter. PFNs4BO: In-context learning for Bayesian optimization. *ICML*, 2023.
- [4] M. Volpp et al. Accurate Bayesian meta-learning by accurate task posterior inference. 2023.
- [5] N. Hansen, A. Auger, R. Ros, O. Mersmann, T. Tušar, and D. Brockhoff. COCO: A platform for comparing continuous optimizers in a black-box setting. *Optimization Methods and Software*, 36(1):114–144, 2021.